# Physics-Informed Neural Network for Satellite Battery Degradation Estimation Based on an Extreme Segment Data Within 2 Min

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***Abstract*—Accurately estimating the state of health (SOH) of satellite batteries is crucial for reliable satellite operation and safe decision-making. However, the complex degradation processes of batteries in space, coupled with limited labeled data, pose challenges for existing SOH estimation methods. We propose a physics-informed neural network (PINN) for satellite battery SOH estimation using only an extreme segment data within 2 min. To address battery degradation complexity, we model degradation attributes from an empirical and state-space perspective. We combine neural networks with physical principles to capture the degradation dynamics of batteries, enabling accurate SOH estimation. To tackle limited and inconsistent data acquisition per cycle, health features are constructed from relaxation voltage data within 2 min, ensuring independence from charging/discharging strategies. Unique factors specific to satellite operations are also considered. To validate the proposed method, the battery degradation experiments simulating a geostationary earth orbit (GEO) satellite are conducted, and an aging dataset is generated. The proposed method achieves a mean absolute percentage error (MAPE) of 1.33% for SOH estimation using a 2-min extreme data segment. Our studies provide valuable insights into battery prognosis and health management for space missions, aiding in optimizing battery performance and longevity in satellite operations.**

***Index Terms*—Health management, lithium-ion battery, physics-informed neural network (PINN), satellite battery, segment data, state-of-health (SOH).**

## I. Introduction

SINCE the dawn of the 21st century, the global satellite application industry has experienced rapid and robust development. As of May 1, 2023, the number of operational satellites in orbit worldwide has reached 7560 [1]. Satellites play an increasingly pivotal role across various domains, including communication, navigation, meteorological monitoring, and Earth observation. Modern society's reliance on satellites continues to escalate, particularly against the backdrop of globalization and the burgeoning information age. Consequently, the reliability and longevity of satellite technology have emerged as critical factors. The successful execution of numerous vital missions is directly contingent upon the seamless operation of satellite systems, necessitating stringent assurance of their dependability.

The power system of a satellite, consisting of solar arrays, battery packs, and power management systems, is crucial for its operational integrity [2]. Solar arrays generate electrical energy during sunlight exposure, while batteries provide backup power in shadowed or suboptimal conditions. Various types of batteries have been developed for satellite power systems, such as nickel-cadmium, nickel-hydrogen, and lithium-ion batteries. Due to their high energy density, and long cycle life, lithium-ion batteries have become the preferred choice for satellite power systems [3]. However, the complex space environment [4], including extreme temperature variations, and radiation, causes battery performance to degrade over time. Consequently, accurately assessing its state of health (SOH) is critical to ensuring the long-term stable operation of satellites.

Battery SOH is a key indicator measuring the current performance of a battery relative to its initial state, reflecting the battery's health during actual operation. Accurate SOH estimation not only aids in predicting the remaining useful life (RUL) of the battery but also supports decision-making for power management systems, ensuring stable power supply throughout the satellite's mission cycle.

Recent advancements in artificial intelligence technology, data-driven methods have dominated battery SOH and RUL estimation [5], [6], [7], [8]. However, a gap exists between terrestrial and spacecraft batteries. Compared with electric vehicles (EVs), satellite battery SOH estimation presents unique challenges. Satellite batteries, with shallow depth of discharge (DOD) [9], generate limited data per cycle, and extreme space conditions complicate data collection and transmission. The few sensors on satellites and restrictions on data frequency and accuracy further limit available data. Published data-driven methods often require complete charge and discharge data,

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using complete charge and discharge curves or extracting statistical features from these curves to estimate SOH, which has high requirements on data quality [10]. Moreover, the degradation process of satellite batteries is influenced by multiple factors, including temperature, discharge rate, duration of shadow periods, DOD, and the number of cycles. These factors vary significantly under different mission and orbital conditions, complicating the development of accurate SOH estimation models. Most existing methods primarily consider the current and voltage curves of the battery charge-discharge process, neglecting other influencing factors. Therefore, it is not feasible to directly apply existing methods for EV to SOH estimation of satellite battery.

To bridge this gap, we propose a physics-informed neural network (PINN) method for satellite battery SOH estimation using only 2-min segment data. Specifically, building upon our previous study [11], we established battery degradation properties from an empirical degradation and state-space perspective and employed neural networks to capture the degradation dynamics of the battery. Previous studies focus on more general SOH estimation. In contrast, this work refines that approach for satellite-specific scenarios, addressing data scarcity and unique operational conditions. By introducing physical knowledge, the data volume requirements can be reduced while achieving more stable and accurate estimates. Considering the unique operational environment of satellites, we extracted features from the relaxation voltage curve within 2 min after the battery was fully charged. Additionally, the factors such as the shadow period, DOD, and calendar aging of the in-orbit satellite are also considered when estimating SOH. To the best of our knowledge, this is the first time that these factors have been considered in satellite battery SOH estimation. The main contributions of this article are as follows.

1) We modeled battery properties from the perspective of empirical degradation and state space. Based on this model, a PINN model is proposed for satellite battery SOH estimation. By integrating neural networks with physical laws and empirical insights, the PINN achieves greater accuracy and reliability, as it adheres to established physical principles while learning from data.
2) To handle inconsistent cycle data due to varying shadow periods, we extracted features from the relaxation voltage data recorded within 2 min after full charge. This ensures the presence of this data segment in every cycle, rendering our method independent of charging/discharging strategies. We also incorporated satellite-specific factors such as DOD, calendar aging, and shadow period duration.
3) We simulated geosynchronous earth orbit (GEO) satellite charging/discharging scenarios to generate a run-to-failure dataset for validation. Results demonstrated that our method effectively estimates SOH using only brief 2-min data segment, achieving promising accuracy.

## II. Related Works

### A. SOH Estimation of Satellite Battery

Following the effective implementation of prognostic and health management (PHM) technology in the F-35 Joint Strike Fighter program [12], PHM systems have been increasingly used in complex systems such as spacecraft and aircraft. Research on spacecraft battery health management and degradation assessment has also grown [13], [14].

Luo et al. [5] proposed a multiscale hierarchical attention bidirectional LSTM with hybrid data preprocessing for spacecraft capacity estimation, validated using the Zhuhai-1 satellite dataset. Cao et al. [15] recognized the mapping relationship between battery discharge cut-off voltage and capacity, and introduced a satellite battery SOH estimation method based on sample entropy. Some other works [16], [17] also extract features from voltage curves to predict battery SOH, avoiding the problem of incomplete charge and discharge of satellite batteries. Considering that satellites usually operate at shallow DOD and roughly consistent currents, Yang et al. [18] proposed a health indicator based on discharge curves to characterize the SOH of the battery. Zhao et al. [2] highlighted the need to consider solar array performance in battery pack health assessments and proposed an Informer-based model to quantify its impact.

### B. Physics-Informed Machine Learning for SOH Estimation

Physics-informed machine learning (PIML) aims to integrate battery aging models with machine learning for better SOH and RUL estimation [19]. Related studies include using empirical models of calendar and cycle combined with LSTM for degradation prediction [20], fusion schemes based on PINN with semiempirical and semiphysical partial differential equations (PDE) [21], and incorporating Fick's diffusion law into neural networks [22]. Some studies use ordinary differential equations (ODE) for degradation modeling [23] or generate synthetic data using physics-based models to train machine learning models [24]. Moreover, some scholars have combined the degradation mechanism during battery aging and considered more internal factors when using machine learning methods to estimate SOH, such as solid electrolyte interface (SEI) growth, loss of positive and negative active materials, and loss of lithium inventory [25].

Nevertheless, these methods usually need complete charge and discharge curves, and do not achieve the deep fusion of physical models and neural networks. How to deeply combine the two remains an open research problem.

## III. Experimental Design and Data Generation

In this study, we develop our experimental plan based on the charging/discharging protocols of a GEO satellite to simulate the aging process of satellite batteries. The experimental platform is shown in Fig. 1. A total of eight batteries were used for aging tests. These batteries, manufactured by LISHEN ($LiNi_{0.5}Co_{0.2}Mn_{0.3}O_2$), have a nominal capacity of 2000 mAh and a nominal voltage of 3.6 V. The cut-off voltages of charging and discharging are 4.2 and 2.5 V.

GEO satellites undergo earth's shadow during the Vernal and Autumnal equinoxes annually. These periods typically occur approximately 23 days before and after the equinoxes, totaling around 46 days each time [26]. This period is referred to as the earth shadow period, while the rest of the time is known as the sunlight period.

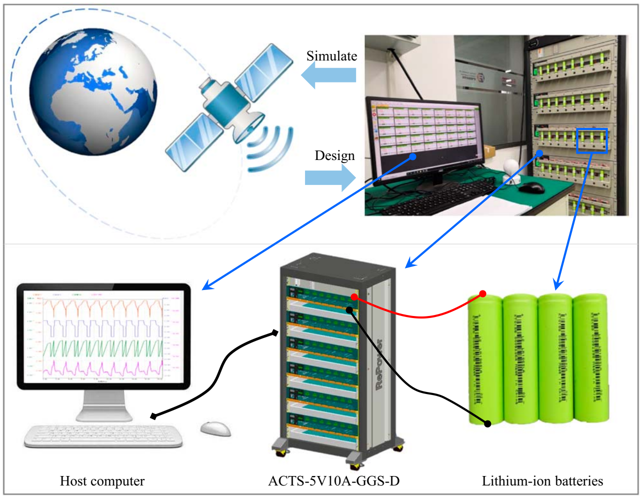


Fig. 1. Experimental platform for simulating satellite battery operation strategies.

TABLE I
DISCHARGE DURATION OF EACH CYCLE OF THE GEO SATELLITE IN THE EARTH'S SHADOW PERIOD [26]

| Cycle number | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 46 | 45 | 44 | 43 | 42 | 41 | 40 | 39 | 38 | 37 | 36 | 35 | 34 | 33 | 32 | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 |
| Duration (min) | 5 | 20 | 34 | 41 | 46 | 50 | 54 | 56 | 58 | 60 | 62 | 64 | 68 | 69 | 70 | 71 | 72 | 72 | 72 | 72 | 72 | 72 | 72 |

In our aging experiments, the batteries are charged to 4.2 V at 2 C and then maintain the voltage unchanged until the current drops to 0.05 C. They are then rest for 5 min to simulate a shelving period. Subsequently, these batteries are discharged at a rate of 0.667 C for a duration as indicated in Table I. Given that the maximum DOD is 80% and the longest shadow period is 1.2 h, the discharge C-rate was set to 0.667 C, ensuring that discharging over 1.2 h results in 80% DOD. Approximately every five cycles, a full discharge is performed to measure the capacity as a reference (discharged at 0.5 C to 2.5 V and recorded the capacity). All batteries were cycled to failure at room temperature.

## IV. METHODOLOGY

This article proposes a PINN method for satellite battery SOH estimation using 2-min segment data, as illustrated in Fig. 2. Key contributions in the offline training stage include feature construction, battery modeling, and PINN development, which are detailed below.

### A. Construction and Selection of Health Features

Robust feature construction significantly improves SOH estimation performance. Various feature extraction methods have been proposed in existing studies tailored to specific datasets and charging/discharging protocols. However, most of these methods are primarily designed for general fully charged and discharged scenarios, rendering them inadequate for the charging/discharging strategies of satellite batteries. Using battery 1 as a case study, we selected health features from consistent data segments across all cycles. Specifically, we analyzed voltage data from the resting phase postfull charge and investigated voltage drops over the same duration at different degradation stages.

Fig. 3 shows the schematic diagram of the feature construction. Following the battery reaching full charge and the disconnection from the power source, there is a slight decline in voltage, as depicted in Fig. 3(b). This voltage is commonly referred to as the relaxation voltage. For the relaxation voltage curve within each cycle, the voltage drop can be computed at uniform time intervals. To comprehensively investigate the correlation between voltage drop ($\Delta V$) and SOH, we sequentially calculate the voltage drop at various start and end times with a minimum resolution of 10 s. Specifically, the time intervals include: [0, 10], [0, 20], [0, 30], ..., [0, 300]; [10, 20], ..., [10, 300]; ..., [290, 300]. For each cycle, the voltage drops within a total of 465 intervals are calculated. Additionally, we have found a correlation between curve entropy and SOH. Therefore, we compute the curve entropy ($S$) within all of the aforementioned intervals using the following equation:

$$S = -\sum_{i=1}^{n} p_i \cdot \log(p_i) \tag{1}$$

where $n$ represents the number of points on the curve, and $p_i$ denotes the normalized value of the curve. After these operations, a total of 930 indicators are obtained. The Pearson correlation coefficient between the voltage drop and curve entropy and SOH are calculated. To address missing or erroneous values in certain intervals for some cycles, we excluded correlations for these intervals from the analysis. The final correlations are illustrated in Fig. 3(f) and 3(g).

The voltage drop shows a negative correlation with SOH, while curve entropy is positively correlated. The strongest correlations are found in intervals starting at 0 s with short duration [Fig. 3(f) and 3(g)]. We analyzed all eight batteries and, given the minor differences between them and indicator redundancy, selected voltage drop and curve entropy features within the intervals [0, 30], [0, 60], [0, 90], and [0, 120] s. These intervals balance computational efficiency and data representativeness, which is critical in satellite applications with limited data sampling rates. The observed trends are consistent with findings in previous studies, where relaxation voltage curves have been shown to effectively reflect battery aging trends [27]. The voltage relaxation curve reflects the redistribution of ions and charge carriers after the battery is disconnected from the power source. Slower relaxation processes or larger voltage drops are usually associated with increased internal resistance, which may be caused by electrode wear or growth of the solid electrolyte interface (SEI) layer [28]. Curve entropy quantifies the complexity of the relaxation voltage curve. Higher entropy may indicate irregular ionic transport or reaction kinetics, which are associated with degradation phenomena such as electrolyte decomposition or electrode structure changes. These features are denoted as $\Delta V_1, \Delta V_2, \Delta V_3, \Delta V_4$ and $S_1, S_2, S_3, S_4$, as illustrated in Fig. 3(c). Fig. 3(d) and 3(e) gives an illustration of the two health features of battery 1.

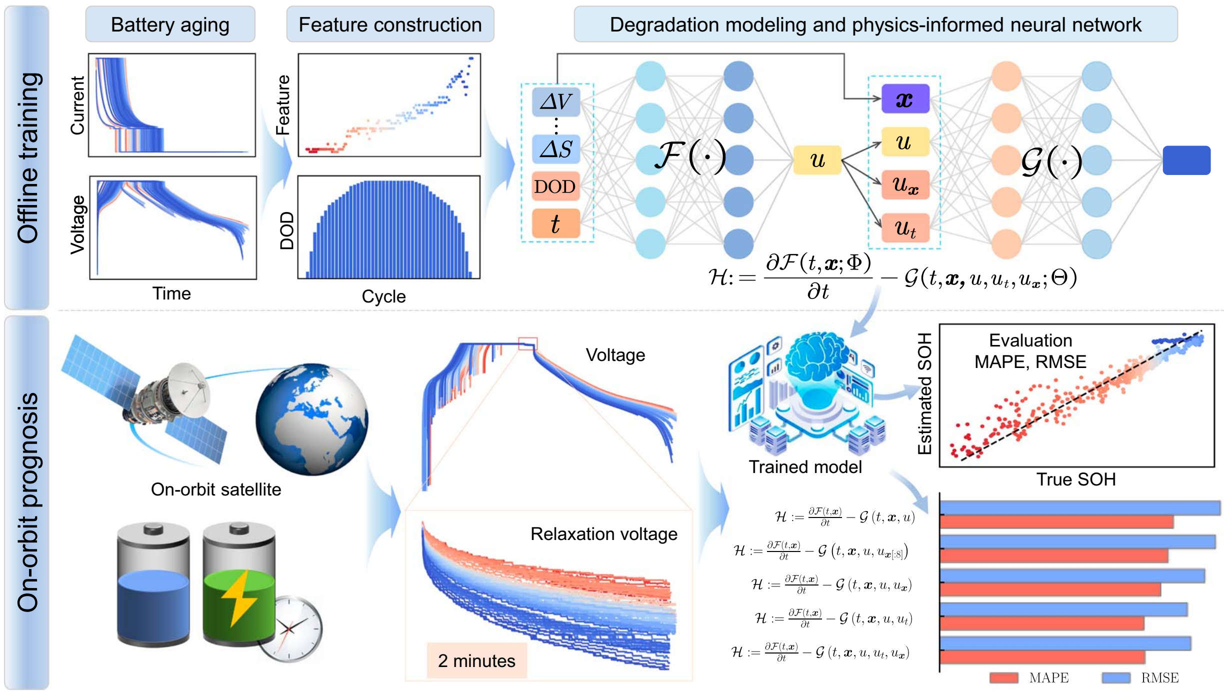


Fig. 2. Overview of the proposed method for satellite battery degradation estimation with 2-min segment data.

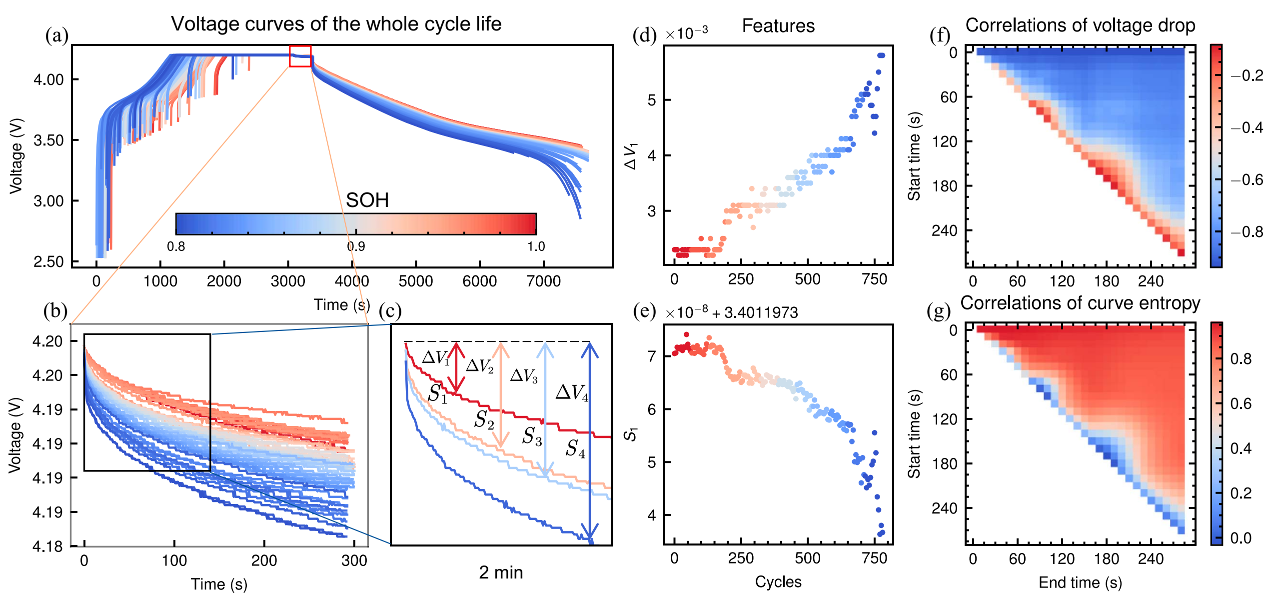


Fig. 3. Diagram of feature extraction. (a) Complete voltage curves. (b) Relaxation voltage curves. (c) Extract features from 2-min segment data. (d) and (e) Visualization of features $\Delta V_1$ and $S_1$. (f) and (g) Correlation heat maps between SOH and voltage drop ($\Delta V$), and SOH and curve entropy ($S$). The x-axis and y-axis represent the start and end time of each interval, respectively. Note the range of colorbar values in the two subfigures. The colder the color in (f), the stronger the negative correlation, and the warmer the color in (g), the stronger the positive correlation.

The aging of batteries is divided into calendar aging and cycle aging [29]. Combined with the unique working mode of GEO satellites, we constructed additional health features based on factors such as the calendar time, shadow period duration, and DOD. Since the constant current discharge mode was used in our experiment, the DOD and shadow period duration are equivalent, thus the DOD (denoted as $F_\delta$) is used as a feature. The shadow period duration of GEO satellites cycles with a period of 46, as shown in Table I. Consequently, features constructed based on the calendar time are also divided into two parts, which we call years indicator and seasonality indicator, represented by $F_\gamma$ and $F_\omega$, respectively. Thus far, a total of 11 health features are constructed, denoted as $\boldsymbol{x} = [\Delta V_1, \Delta V_2, \Delta V_3, \Delta V_4, S_1, S_2, S_3, S_4, F_\delta, F_\gamma, F_\omega]$.

### B. Proposed Physics-Informed Neural Network for SOH Estimation

*1) Battery Degradation Modeling:* The performance of battery aging is that the available capacity decays and the internal resistance increases. SOH can be used to reflect the aging level of a battery. From the perspective of capacity degradation, SOH can be calculated as

$$u^k = \frac{Q^k}{Q^0} \tag{2}$$

where $u^k$ represents the SOH of the cycle $k$, $Q^k$ is the capacity of cycle $k$, and $Q^0$ is the nominal capacity. If we use a continuous function $u(t)$ to represent the degradation trajectory of the battery over time $t$, then the SOH over its entire lifespan is the point on the trajectory when $t = k \ \ (k = 1, 2, \cdots, n)$, that is $u^k = u(k)$.

To accurately predict the SOH and RUL of batteries, various empirical models have been proposed. These models, including the linear model, exponential model, and power-law model [30], characterize the degradation of battery capacity over time or cycle. They all represent the battery's deterioration trajectory as a function dependent on a single variable, either time or cycle.

However, simplifying battery degradation solely as a function of time overlooks other crucial factors. Battery degradation is influenced not just by time, but also by charging and discharging rates, calendar time, temperature, DOD, etc. [29]. Therefore, we propose a multivariable function to model the aging trajectory of satellite batteries

$$u = f(t, \boldsymbol{x}) \tag{3}$$

where $t$ represents time and $\boldsymbol{x}$ represents a vector composed of DOD, shadow duration, calendar time, discharging rate, temperature, and health features. In this work, considering our experiments, $\boldsymbol{x}$ consists of DOD, calendar time, and health features.

Without loss of generality, to describe the degradation dynamics of the satellite battery, its SOH decay rate can be describe as

$$\frac{\partial f\left(t, \boldsymbol{x}\right)}{\partial t} = g\left(t, \boldsymbol{x}, u; \theta\right). \tag{4}$$

The (4) constitutes an explicit PDE defined by parameters $\theta$, with $g(\cdot)$ denoting a nonlinear function of $t$, $\boldsymbol{x}$, and $u$. This dynamic function $g(\cdot)$ encapsulates the internal degradation mechanisms within the battery, and modifying this nonlinear function enables the representation of different degradation patterns. Specifically, when solely considering time, models such as the linear, exponential, power-law, and failure forecast models can be regarded as specific instances of (4).

*2) Physics-Informed Neural Network Based on Degradation Model:* According to (4), we define a physics-informed neural network for satellite battery degradation modeling

$$\mathcal{H} := u_t - g\left(t, \boldsymbol{x}, u; \theta\right) \tag{5}$$

where $u_t = (\partial u/\partial t) = (\partial f\left(t, \boldsymbol{x}\right)/\partial t)$. A neural network, denoted as $\mathcal{F}(t, \boldsymbol{x}; \Phi)$ with learnable parameters $\Phi$, is used to model $f(t, \boldsymbol{x})$. We called it solution neural network. Thus, $u_t$ can be calculated as $(\partial \mathcal{F}\left(t, \boldsymbol{x}; \Phi\right)/\partial t)$ with automatic differentiation mechanisms of deep learning. An unavoidable challenge is that the explicit form of dynamic $g(\cdot)$ is unknown. Inspired by [31], a neural network $\mathcal{G}(\cdot)$ with learnable parameters $\Theta$ (called dynamic neural network) is introduced to approximate $g(\cdot)$ to modeling battery dynamics. Therefore, the physics-informed neural network given in (5) for battery degradation is reformulated as

$$\mathcal{H} := \frac{\partial \mathcal{F}\left(t, \boldsymbol{x}; \Phi\right)}{\partial t} - \mathcal{G}\left(t, \boldsymbol{x}, u, u_t, u_{\boldsymbol{x}}; \Theta\right) \tag{6}$$

where $u_{\boldsymbol{x}} = [(\partial u/\partial x_1), (\partial u/\partial x_2), \cdots]^\top$ represents the first-order partial derivative of $u$ with respect to $\boldsymbol{x}$. The Dynamic neural network $\mathcal{G}(\cdot)$ propose a more flexible relationship to $t$, $\boldsymbol{x}$, $u$, and their first-order partial derivative, allowing the model to learn more essential battery degradation dynamics.

Equation (6) is derived from (4). However, due to the neural network's discrete training process, it does not strictly satisfy (4). Consequently, it is necessary to ensure that $\mathcal{H}(t, \boldsymbol{x}) = 0$ at any discrete sample point to approximate (4). That is, the optimization process for the proposed PINN should conform to the PDE loss outlined by (4), specifically

$$\mathcal{L}_{\text{PDE}} = \sum_{i=1}^{N} \left|\mathcal{H}\left(t^i, \boldsymbol{x}^i\right)\right|^2 \tag{7}$$

where superscript $i$ represents the $i$th sample and $N$ is the number of samples. In addition, the optimization objective also includes data fitting loss, that is, mean squared error (MSE) loss

$$\mathcal{L}_{\text{MSE}} = \sum_{i=1}^{N} \left|u^i - \hat{u}^i\right|^2 \tag{8}$$

where $u^i$ and $\hat{u}^i$ represent the true and predicted SOH. The total function is formulated as

$$\mathcal{L} = \mathcal{L}_{\text{MSE}} + \alpha \mathcal{L}_{\text{PDE}} \tag{9}$$

where the $\alpha$ is trade-off parameter.

### C. Implementation Details

The structure of the proposed PINN is depicted in the upper right of Fig. 2. Both the solution network $\mathcal{F}(\cdot)$ and dynamic network $\mathcal{G}(\cdot)$ are simple multilayer perceptrons. Our model was optimized by minimizing the loss in (9), with the parameter $\alpha = 0.1$ determined through grid search. The Adam optimizer is employed during the training process, the batch size is set to 512, the epoch is set to 200, and the early stopping is set to 20 epochs. The dataset consists of data from eight batteries. We adopt a leave-one-out cross-validation approach. In each experiment, data from seven batteries are used to train the model, and the remaining battery is used as the test set. This ensures that the model is tested on entirely unseen data in each fold.

To evaluate the PINN's performance, we used mean absolute percentage error (MAPE) and root mean squared error (RMSE), defined as

$$\text{MAPE} = \frac{1}{N} \sum_{i=1}^{N} \left|\frac{u^i - \hat{u}^i}{u^i}\right| \times 100 \tag{10}$$

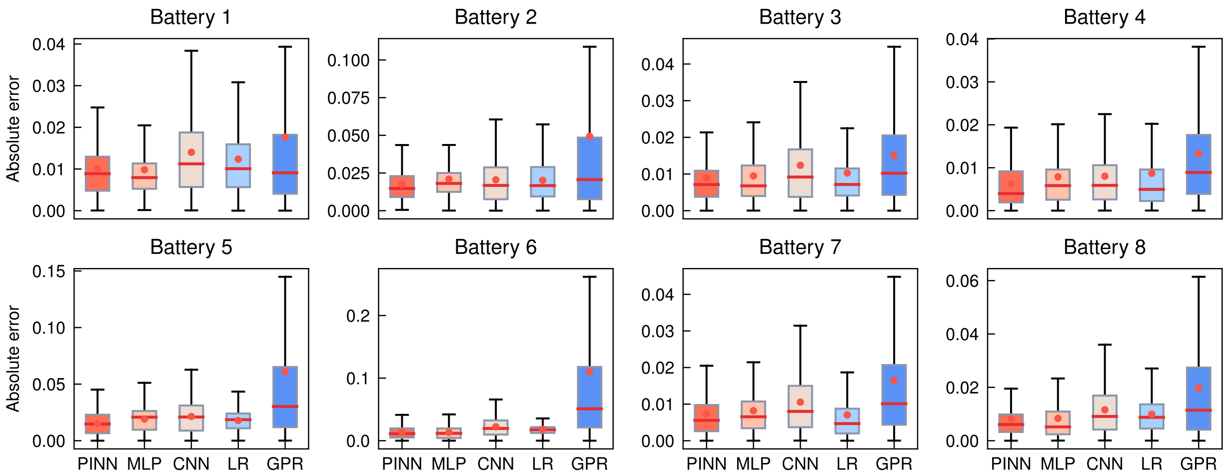


Fig. 4. Boxplots for all batteries and all methods. The y-axis is the absolute error between the predicted value and true value. The red dot represents the mean of 10 experiments, and the red line represents the median. Smaller boxes represent methods with smaller errors.

$$\mathrm{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(u^i - \hat{u}^i)^2} \tag{11}$$

where $u^i$ and $\hat{u}^i$ represent the true and estimated SOH of the sample $i$, and $N$ denotes the number of test samples.

## V. Experimental Results and Discussion

### A. Comparison With Typical Method

To verify the superiority of the proposed PINN structure, we compare it against four other prominent data-driven methods. These methods include GPR, linear regression (LR), multilayer perceptron (MLP), and convolutional neural network (CNN). GPR and LR are well-known shallow learning methods frequently used in battery SOH estimation [32], [33], while MLP and CNN represent classic neural network models, detailed in Table II. The MLP shares the same structure and parameter count as $\mathcal{F}(\cdot)$ in our PINN. Additionally, the CNN employs a skip connection structure, maintaining a similar order of magnitude in parameters as the PINN.

The SOH estimation results are provided in Table III, revealing insightful findings about the effectiveness of various methodologies. Notably, our proposed method demonstrates exceptional accuracy across most batteries, underscoring its robustness in prognosticating battery degradation. Examining the aggregate performance across all batteries, our method emerges as the frontrunner, boasting the lowest average estimation errors. This distinction is particularly evident when considering the MAPE and RMSE, which stand at 1.331% and 1.570%, respectively. These metrics serve as vital benchmarks for evaluating the reliability and precision of SOH estimation techniques.

To further analyze estimation performance at an individual cycle level, we present scatter plots of predicted SOH versus ground truth and error distribution plots in Fig. 5. These visualizations illustrate the model's prediction performance across all cycles and complement the global evaluation metrics.

A key distinguishing factor of our proposed approach lies in its utilization of the dynamic network $\mathcal{G}(\cdot)$ to capture the intricate degradation dynamics inherent to battery systems. While both the MLP and our proposed PINN have an identical forward inference module, it is the incorporation of $\mathcal{G}(\cdot)$ that empowers the PINN to achieve superior accuracy. By leveraging this advanced modeling framework, our method effectively

TABLE II
Details of the Proposed PINN, MLP, and CNN Methods

| Model | Module | Layer | Input dim | Output dim | Dropout | Para. Num |
|---|---|---|---|---|---|---|
| PINN | $\mathcal{F}(\cdot)$ | Linear+Sin | 11 | 50 | 0.1 | 2501 |
| | | Linear+Sin | 50 | 32 | | |
| | | Linear+ReLU | 32 | 32 | | |
| | | Linear | 32 | 1 | | |
| | $\mathcal{G}(\cdot)$ | Linear+Sin | 23 | 100 | | |
| | | Linear+Sin | 100 | 1 | | |
| MLP | | Linear+Sin | 11 | 50 | 0.1 | 2501 |
| | | Linear+Sin | 50 | 32 | | |
| | | Linear+ReLU | 32 | 32 | | |
| | | Linear | 32 | 1 | | |
| CNN | | ResBlock | (1, 12) | (12, 6) | | 2921 |
| | | ResBlock | (12, 6) | (12, 3) | | |
| | | Linear+ReLU | 36 | 32 | | |
| | | Linear | 32 | 1 | | |

Note: The MLP has the same structure and parameter amount as Solution network $\mathcal{F}(\cdot)$ in PINN. The last column represents the number of parameters for the forward inference process. After the model is trained, Dynamic network $\mathcal{G}(\cdot)$ does not participate in the inference process, so its parameters are not included in the statistics.

TABLE III
SOH Estimation Errors (%) of the Proposed PINN and Other Four Methods

| | PINN | | MLP | | CNN | | GPR | | LR | |
|---|---|---|---|---|---|---|---|---|---|---|
| Battery | MAPE | RMSE | MAPE | RMSE | MAPE | RMSE | MAPE | RMSE | MAPE | RMSE |
| 1 | **1.102** | **1.349** | 1.195 | 1.429 | 1.474 | 1.843 | 1.967 | 3.701 | 1.343 | 1.605 |
| 2 | **2.093** | **2.284** | 2.276 | 2.545 | 2.454 | 3.022 | 5.561 | 11.168 | 2.190 | 2.453 |
| 3 | **1.134** | **1.462** | 1.296 | 1.619 | 1.336 | 1.764 | 1.657 | 2.191 | 1.136 | 1.579 |
| 4 | **0.845** | **1.078** | 0.882 | 1.133 | 1.272 | 1.583 | 1.451 | 2.041 | 0.968 | 1.380 |
| 5 | **1.808** | **1.977** | 1.888 | 2.082 | 2.348 | 2.686 | 6.927 | 10.925 | 1.955 | 2.020 |
| 6 | **1.757** | **2.013** | 2.254 | 2.501 | 3.231 | 3.782 | 11.262 | 18.376 | 1.941 | 2.030 |
| 7 | 0.927 | 1.143 | 1.062 | 1.269 | 1.246 | 1.547 | 1.805 | 2.645 | **0.770** | **1.096** |
| 8 | **0.985** | 1.253 | 1.382 | 1.662 | 1.465 | 1.801 | 2.163 | 3.001 | 1.066 | **1.230** |
| Aver. | **1.331** | **1.570** | 1.529 | 1.780 | 1.853 | 2.253 | 4.099 | 6.756 | 1.421 | 1.674 |

Note: All values in the table are averaged from 10 experiments. The best results have been bolded.

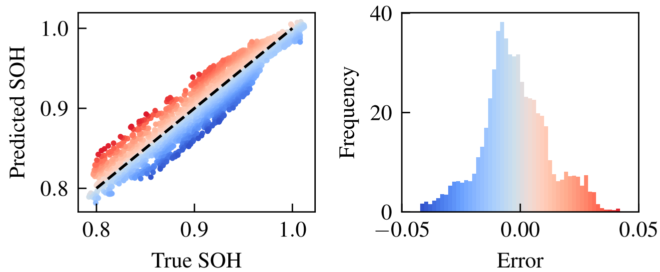


Fig. 5. Comparison of predicted SOH and ground truth values (left) and error distribution across all test cycles (right).

encapsulates the nuanced interplay of factors influencing battery health, thereby enhancing the fidelity of SOH estimations. These results underscore the pivotal role of model architecture and physical knowledge in the domain of battery prognosis and health management. Beyond mere predictive performance, the ability to discern and represent underlying physical processes is crucial for ensuring the reliability of prognostic models. As such, our findings not only validate the efficacy of the proposed PINN but also highlight the significance of adopting a physics-informed approach for addressing the complex dynamics inherent to battery degradation.

To provide a more visually insightful depiction of the performance disparity among the various methods, we constructed error boxplots showcasing the distribution of estimation errors across all batteries and all cycles, as illustrated in Fig. 4. This visualization offers a comprehensive overview of the variability and consistency in estimation accuracy exhibited by each method.

Notably, our proposed PINN exhibits superior performance across multiple dimensions. The distribution of estimation errors for the PINN demonstrates tighter clustering around the mean and median values, indicative of its heightened precision and reliability. In contrast, the MLP and LR models exhibit marginally larger error spreads, albeit still maintaining commendable accuracy levels. Conversely, the GPR model registers the widest error distribution, signaling comparatively lower accuracy and greater variability in estimation outcomes. Overall, Fig. 4 demonstrates the advantage of PINN in terms of prediction stability, as the variance of its prediction error is significantly smaller compared with other methods. This stability highlights the robustness of PINN, which leverages physical insights to mitigate overfitting and ensure consistent performance.

Drawing inference from these observations, several noteworthy conclusions can be gleaned. First, the efficacy of the designed features is underscored by the satisfactory accuracy achieved across the most methods. Despite utilizing only 2-min data segment, our models adeptly capture the salient attributes indicative of battery health, attesting to the discriminative power of the designed features. Second, the inherent structural advantages embedded within our proposed PINN architecture emerge as a pivotal determinant of superior performance. By adeptly integrating the dynamic properties governing battery degradation, the PINN model transcends the limitations of conventional data-driven approaches, thereby achieving remarkable estimation accuracy. This nuanced consideration of degradation dynamics not only enhances predictive fidelity but also imbues the model with a deeper understanding of the underlying physical processes governing battery health evolution.

In essence, these findings underscore the multifaceted advantages conferred by our proposed methodology, affirming its efficacy in the realm of SOH estimation. Beyond mere predictive performance, the holistic integration of feature engineering and model architecture optimization underscores a paradigm shift in battery health management toward predictive frameworks that incorporate more physical knowledge.

To evaluate the efficiency of the proposed method, we measured the average prediction time during testing. The model contains 2501 parameters during inference, as reported in Table II. The inference time for a single sample was evaluated on a platform using Pytorch 1.7.1, running on an Intel(R) Core(TM) i5-10400F CPU @ 2.90GHz. Based on 10 test runs, the average prediction time per sample was 1.17e-5 s. These results demonstrate the method's high computational efficiency, making it feasible for real-world applications.

## B. Ablation Studies

To further explore the effectiveness of the extracted features and the advancement of the PINN structure, we carried out ablation experiments and conducted further analysis.

*1) Ablation Study on Designed Features:* In this study, we proposed to extract health features from 2-min segment data. Combined with the unique shadow period of GEO satellites, we designed a comprehensive set of input features $\boldsymbol{x} = [\Delta V_1, \Delta V_2, \Delta V_3, \Delta V_4, S_1, S_2, S_3, S_4, F_\delta, F_\gamma, F_\omega]$, which integrates the DOD, calendar time, and health-related features to enhance the SOH estimation.

To rigorously evaluate the impact of the DOD and calendar time on model performance, we divided $\boldsymbol{x}$ into different subsets and used them as inputs to the proposed PINN to estimate SOH. The results, presented in Table IV and Fig. 6, provide compelling evidence of the efficacy of our designed features.

It can be seen from the table that "with all" achieved the best results on all batteries except battery 5. Fig. 6 offers a more intuitive visualization, highlighting the benefits of including DOD and calendar time in the model inputs. In most scenarios, the "with all" feature set yielded better results compared with subsets that excluded $F_\delta$, which in turn better than the results without $F_\delta$, $F_\gamma$, and $F_\omega$. This trend illustrates the critical role that these features play in capturing the nuanced dynamics of battery degradation, and also demonstrates the usefulness of considering DOD and calendar time in estimating the SOH of satellite batteries.

*2) Ablation Study on Proposed PINN Structure:* In this section, we evaluated the estimation errors of the proposed PINN as well as its variants on battery 1. The proposed PINN is formulated as (6), where we incorporate the first-order partial derivatives of $u$ with respect to input features $\boldsymbol{x}$ and time $t$ when modeling the degradation dynamics. To comprehensively assess the impact of these derivatives, we designed four variations of

TABLE IV
SOH ESTIMATION ERRORS (%) OF THE PROPOSED PINN WITH DIFFERENT INPUTS

| Battery | With All MAPE | With All RMSE | Without $F_\delta$ MAPE | Without $F_\delta$ RMSE | Without $F_\delta, F_\gamma, F_\omega$ MAPE | Without $F_\delta, F_\gamma, F_\omega$ RMSE |
|---|---|---|---|---|---|---|
| 1 | **1.102** | **1.349** | 1.408 | 1.643 | 1.309 | 1.559 |
| 2 | **2.093** | **2.284** | 2.435 | 2.751 | 2.401 | 2.706 |
| 3 | **1.134** | **1.462** | 1.250 | 1.529 | 1.335 | 1.630 |
| 4 | **0.845** | **1.078** | 0.900 | 1.120 | 1.206 | 1.428 |
| 5 | 1.808 | 1.977 | 1.727 | 1.928 | **1.717** | **1.886** |
| 6 | **1.757** | 2.013 | 2.024 | 2.273 | 1.788 | **2.000** |
| 7 | **0.927** | **1.143** | 1.122 | 1.331 | 1.416 | 1.666 |
| 8 | **0.985** | **1.253** | 1.191 | 1.449 | 1.335 | 1.592 |
| Aver. | **1.331** | **1.570** | 1.507 | 1.753 | 1.563 | 1.808 |

Note: All values in the table are averaged from 10 experiments. The best results have been **bolded**. "with all" means that the proposed PINN takes $\boldsymbol{x} = [\Delta V_1, \Delta V_2, \Delta V_3, \Delta V_4, S_1, S_2, S_3, S_4, F_\delta, F_\gamma, F_\omega]$ as input to estimate SOH. The other two cases indicate that the corresponding item is removed from the input $\boldsymbol{x}$.

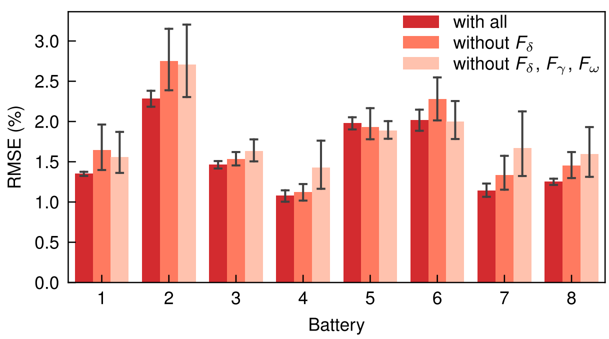


Fig. 6. SOH estimation errors (%) of the proposed PINN with different inputs. "with all" means that the proposed PINN takes $\boldsymbol{x} = [\Delta V_1, \Delta V_2, \Delta V_3, \Delta V_4, S_1, S_2, S_3, S_4, F_\delta, F_\gamma, F_\omega]$ as input to estimate SOH. The other two cases indicate that the corresponding item is removed from the input $\boldsymbol{x}$.

the PINN. Each variant omits certain derivatives to isolate their effects on estimation accuracy.

1) *No Derivatives:* This variant does not consider any partial derivatives of $u$, relying solely on the base input features.
2) *Temporal Derivative Only:* This variant only considers the partial derivative of $u$ with respect to time $t$, capturing the temporal aspect of degradation.
3) *Feature Derivatives Only:* This variant only considers the partial derivatives of $u$ with respect to the input features $\boldsymbol{x}$, focusing on feature-driven degradation analysis.
4) *Limited Feature Derivatives:* This variant includes only the first eight partial derivatives of $u$ with respect to the input features $\boldsymbol{x}$, to evaluate the impact of health features derivatives on model performance.

The formulas and evaluation results of these variants are shown in Fig. 7. By evaluating these variants, we aim to determine the contribution of each type of derivative to the overall accuracy of the SOH estimation.

From the results in the Fig. 7, it is evident that the inclusion of different derivative terms affects the performance of the model. Specifically, the variant that includes any derivative terms performs slightly better than the variant that does not consider derivative terms. This suggests that both temporal and feature derivatives contribute positively to the accuracy of the model.

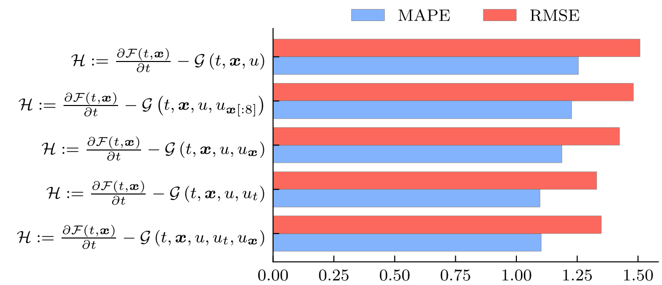


Fig. 7. SOH estimation errors (%) of the proposed PINN (6) and its variants on battery 1. $u_t$ represents the derivative of $u$ with respect to $t$, $u_{\boldsymbol{x}}$ represents the derivative of $u$ with respect to $\boldsymbol{x}$, and $u_{\boldsymbol{x}[:8]}$ represents the derivative of $u$ with respect to the first eight terms in $\boldsymbol{x}$, that is, $\boldsymbol{x}[:8] = [\Delta V_1, \Delta V_2, \Delta V_3, \Delta V_4, S_1, S_2, S_3, S_4]$.

To further elaborate, the incorporation of temporal derivatives helps the model capture the evolution of battery degradation over time. On the other hand, feature derivatives provide insight into how specific operational conditions and usage patterns impact the degradation process. For instance, variables such as DOD, calendar time, and cycle numbers are critical in modeling the complex dynamics within the battery.

## VI. CONCLUSION

We developed a physics-informed neural network for estimating satellite battery SOH using only an extreme 2-min data segment. This method addresses the challenges of complex degradation in the harsh space environment and limited data availability. By integrating physical principles into the neural network, the model effectively captures degradation dynamics. The PINN utilizes relaxation voltage data, making it independent of charging/discharging strategies and enhancing its practicality. Additionally, factors such as DOD, calendar aging, and shadow period duration are considered to boost accuracy and robustness.

Compared with our previous work, which focused on general SOH estimation using charging data across diverse datasets, this study refines the approach for satellite-specific scenarios. By leveraging only relaxation voltage data, our method addresses the data-scarce conditions typical of satellite applications. Furthermore, the satellite-specific features designed in this work, such as DOD and shadow period duration, were shown through ablation studies to significantly enhance SOH estimation accuracy.

Despite its contributions, this study has certain limitations. First, the proposed method was validated on simulated satellite battery data, which may not fully capture the complexity and variability of real-world operational conditions. Second, while the PINN framework integrates physical insights into the neural network, the current implementation still relies on manually designed constraints. A deeper fusion of neural networks with physical models, such as learning physics-informed differential operators or adaptive constraint weighting, could further enhance the model's interpretability and generalizability. Third, this study was conducted under room temperature conditions, while real satellite environments experience temperature fluctuations that can affect battery relaxation voltage and SOH estimation. Temperature variations affect internal battery dynamics, including ion transport, SEI layer growth, and electrolyte behavior, leading to different relaxation characteristics. These limitations highlight areas where further validation and methodological improvements are needed.

Future research could apply this method to various battery types, explore other orbital conditions, and different operating temperatures to improve estimation accuracy. Additionally, exploring alternative approaches for integrating physics more dynamically within the neural network could enhance the method's applicability. These directions provide promising pathways for advancing satellite battery SOH monitoring and mission reliability.

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